INDUSTRY PROBLEM SOLVING TASK

1. Autonomous Vehicle Vision System Selection

An automobile company is developing an autonomous vehicle detection system that must operate under the following conditions:

- If accuracy < 90% → system fails
- If latency > 70 ms → unacceptable
- The vehicle supports moderate computational resources

Methods available:

- Viola-Jones: 82% accuracy, 30 ms latency, low computation
- YOLO: 93% accuracy, 60 ms latency, moderate computation
- Deep Learning: 97% accuracy, 90 ms latency, high computation

Task:
Compare the three methods and evaluate which method satisfies all the constraints. Justify your decision using logical reasoning based on accuracy, latency, and computational requirements.

2. Smart Factory Product Inspection Selection

A manufacturing company wants to deploy a real-time computer vision system to detect defective products on a production line. The system must operate under the following conditions:

- If accuracy < 95% → defects may be missed
- If processing time > 50 ms → production is affected
- Hardware supports moderate computation

Methods available:

- Traditional Image Analysis: 91% accuracy, 25 ms, low computation
- YOLO: 96% accuracy, 45 ms, moderate computation
- Deep Learning: 98% accuracy, 75 ms, high computation

Task:
Compare the three methods and evaluate which method satisfies all the constraints. Justify your decision using logical reasoning based on accuracy, processing time, and computational requirements.

1. Autonomous Vehicle Vision System Selection

Problem Understanding

An automobile company wants to develop an autonomous vehicle vision system. The system must detect objects such as vehicles, pedestrians, traffic signs, and obstacles accurately and quickly.

The system has three important constraints:

1. Accuracy must be at least 90%.
   - If accuracy is below 90%, the system fails.
2. Latency must not exceed 70 ms.
   - If latency is greater than 70 ms, the response is too slow for an autonomous vehicle.
3. Computational requirement must be moderate or lower.
   - The vehicle has only moderate computational resources.

Therefore, a method is acceptable only if it satisfies all three conditions simultaneously.

Comparison of Methods

Method	Accuracy	Latency	Computation	Accuracy Requirement	Latency Requirement	Computation Requirement	Overall
Viola-Jones	82%	30 ms	Low	✖ Fail	☑ Pass	☑ Pass	✖ Reject
YOLO	93%	60 ms	Moderate	☑ Pass	☑ Pass	☑ Pass	☑ Select
Deep Learning	97%	90 ms	High	☑ Pass	✖ Fail	✖ Fail	✖ Reject

A. Viola-Jones

Accuracy = 82%

The minimum required accuracy is 90%.

Therefore:

82% < 90% → Fails

Although Viola-Jones has some advantages:

- Latency is only 30 ms, so it is very fast.
- It requires low computation.
- It can work well on systems with limited hardware.

However, speed alone is not sufficient for an autonomous vehicle. An accuracy of 82% means the system could incorrectly detect or miss important objects.

Result: Reject Viola-Jones.

B. YOLO

Accuracy = 93%

Required accuracy = 90%.

Therefore:

93% ≥ 90% → Pass

Latency = 60 ms

Maximum allowed latency = 70 ms.

Therefore:

60 ms ≤ 70 ms → Pass

Computation = Moderate

Available computational resources = Moderate.
Therefore:
Moderate computation ≤ Moderate available resources → Pass
YOLO satisfies every requirement:

- Accuracy → Pass
- Latency → Pass
- Computation → Pass

Result: Select YOLO.

C. Deep Learning

Accuracy = 97%
Required accuracy = 90%.
Therefore:
97% ≥ 90% → Pass
However:
Latency = 90 ms
Maximum allowed latency = 70 ms.
Therefore:
90 ms > 70 ms → Fail
It also requires high computation, while the vehicle supports only moderate computational resources.
Therefore:
High computation > Moderate available computation → Fail
Although Deep Learning provides the highest accuracy, it cannot satisfy the real-time and hardware requirements.
Result: Reject Deep Learning.

Logical Decision

The selection condition is:
Accuracy ≥ 90% AND Latency ≤ 70 ms AND Computation ≤ Moderate
For YOLO:
93% ≥ 90% → TRUE
60 ms ≤ 70 ms → TRUE
Moderate ≤ Moderate → TRUE
Therefore:
TRUE AND TRUE AND TRUE = TRUE

Final Decision

YOLO is the most suitable method for the autonomous vehicle vision system.
It provides a good balance between accuracy, speed, and computational requirements. Viola-Jones is faster but not accurate enough, while Deep Learning is more accurate but too slow and computationally expensive.

Conclusion

For an autonomous vehicle, accuracy and response time are both critical. A highly accurate system that responds too slowly may still be unsafe, while a fast system with poor accuracy may miss important objects. YOLO provides the best balance and satisfies all the given constraints.

2. Smart Factory Product Inspection Selection
Problem Understanding
A manufacturing company wants to use computer vision to automatically identify defective products on a production line.
The system must satisfy three conditions:
1. Accuracy must be at least 95%.
   - Accuracy below 95% may cause defective products to be missed.
2. Processing time must not exceed 50 ms.
   - Higher processing time can slow down the production line.
3. The hardware supports moderate computation.
   - Therefore, methods requiring high computational power are unsuitable.

The selected method must satisfy all three conditions.
Comparison of Methods

Method	Accuracy	Processing Time	Computation	Accuracy ≥ 95%	Time ≤ 50 ms	Computation Suitable?	Overall
Traditional Image Analysis	91%	25 ms	Low	❌ Fail	☑ Pass	☑ Pass	❌ Reject
YOLO	96%	45 ms	Moderate	☑ Pass	☑ Pass	☑ Pass	☑ Select
Deep Learning	98%	75 ms	High	☑ Pass	❌ Fail	❌ Fail	❌ Reject

A. Traditional Image Analysis
Accuracy = 91%
Required accuracy = 95%.
Therefore:
91% < 95% → Fail
The method is very fast:
25 ms ≤ 50 ms → Pass
It also requires low computation, which is suitable for the hardware.
However, the accuracy requirement is mandatory. If the system misses defective products, it could allow poor-quality products to continue through the production process.
Therefore, despite its speed and low computational requirements:
Result: Reject Traditional Image Analysis.
B. YOLO
Accuracy = 96%
Required accuracy = 95%.
Therefore:
96% ≥ 95% → Pass
Processing time = 45 ms
Maximum allowed processing time = 50 ms.
Therefore:
45 ms ≤ 50 ms → Pass
Computation = Moderate

Available computation = Moderate.
Therefore:
Moderate computation ≤ Moderate available computation → Pass
Thus, YOLO satisfies all three requirements:

- Accuracy → 96% → Pass
- Processing time → 45 ms → Pass
- Computation → Moderate → Pass

Result: Select YOLO.

C. Deep Learning

Accuracy = 98%
Required accuracy = 95%.
Therefore:
98% ≥ 95% → Pass
Deep Learning provides excellent accuracy and would be very effective for detecting defects.
However:
Processing time = 75 ms
Maximum allowed = 50 ms.
Therefore:
75 ms > 50 ms → Fail
It also requires high computation, while the available hardware supports only moderate computation.
Therefore:
High computation > Moderate computation → Fail
Even though its accuracy is the highest, it cannot be selected because it violates the real-time processing and hardware constraints.
Result: Reject Deep Learning.

Logical Decision

The selection condition is:
Accuracy ≥ 95% AND Processing Time ≤ 50 ms AND Computation ≤ Moderate
For YOLO:
96% ≥ 95% → TRUE
45 ms ≤ 50 ms → TRUE
Moderate ≤ Moderate → TRUE
Therefore:
TRUE AND TRUE AND TRUE = TRUE

Final Decision

YOLO is the most suitable method for the smart factory product inspection system.
It provides:

- 96% accuracy, which is sufficient to minimize missed defects.
- 45 ms processing time, which is fast enough for the production line.
- Moderate computation, which matches the available hardware.

Conclusion

Traditional Image Analysis is fast and computationally efficient, but its 91% accuracy is insufficient. Deep Learning provides the best accuracy at 98%, but its 75 ms processing time and high computational requirement make it unsuitable. YOLO offers the best overall balance, satisfying accuracy, speed, and hardware requirements.